CS445 LAB ASSIGNMENT 5

Campus Event Board — Notes

Section A: Design System & Box Model | Section B: CSS Grid & Responsive Design

Team Members

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Answers

1. Show your :root palette. Why are named variables better than repeating hex codes across the file?

The project uses CSS custom properties in :root:
--brand: #2b547e;
--brand-light: #e5eef5;
--ink: #333;
--paper: #fff;
--link: #164a78;
--visited-link: #5c2d7d;
--active-link: #9b2c2c;

Named variables allow the same design values to be reused consistently. A colour can be changed once in :root and the change is reflected everywhere that uses var(--…). This improves consistency and maintainability.

2. Give one use each of hex, rgb() and hsl() from your CSS, and say where you used a semi-transparent colour and why.

Hex is used for the brand colour, for example #2b547e. rgb() is used for a semi-transparent glass/card colour such as rgb(255 255 255 / 35%). hsl() is used for an accent colour, for example hsl(210 55% 35%). The semi-transparent colour is used on the glass-effect event cards and information panels so the background remains partially visible and creates the frosted-glass appearance.

3. Why prefer rem over px for fonts and spacing? Show one calc() and one clamp() you used and what each does.

rem is preferred for most fonts and spacing because it scales from the root font size and is more adaptable to user/browser text-size settings. px is useful for very thin borders.

calc() example: width: calc(100% - 2rem); — calculates the available width minus 2rem.

clamp() example: font-size: clamp(1.5rem, 4vw, 3rem); — makes the heading fluid while keeping it between 1.5rem and 3rem.

4. A box has width: 200px; padding: 20px; border: 5px. What is its actual rendered width with content-box, and with border-box? Which did you use and why?

With content-box: 200 + 20 + 20 + 5 + 5 = 250px. With border-box, the declared 200px includes content, padding and border, so the rendered width is 200px. The project uses border-box because width calculations become predictable and padding/borders do not unexpectedly increase the total size.

5. Name the four layers of the box model, from the inside out.

1. Content
2. Padding
3. Border
4. Margin

6. What does repeat(auto-fit, minmax(15rem, 1fr)) do, and why does it need no media query?

It creates as many columns as can fit in the available space. Each card has a minimum width of 15rem and can expand to one equal fraction (1fr). auto-fit automatically changes the number of columns as the available width changes, so a media query is not needed for this card-grid reflow.

7. Mobile-first: why write the phone styles as the base and add layout with min-width queries? And why is the viewport <meta> tag essential?

Mobile-first starts with a simple single-column layout for small screens and progressively adds columns and larger layouts with min-width media queries. This keeps the base CSS simple and provides progressive enhancement. The viewport meta tag makes the browser use the device width and correct initial scale, which is essential for responsive layouts on phones.

8. Paste your grid-template-areas map. Which regions span more than one column, and how did you make them?

Example desktop map:

```
grid-template-areas:
  "header header"
  "nav nav"
  "main sidebar"
  "footer footer";
```

Header, navigation and footer span two columns because their area names are repeated across both columns. Main and sidebar occupy separate columns. The mobile layout changes to a single-column map so the regions stack vertically.

9. Name one responsive technique for images and one for type that need no media query.

Images: max-width: 100%; height: auto; so they scale down with their container without overflowing.

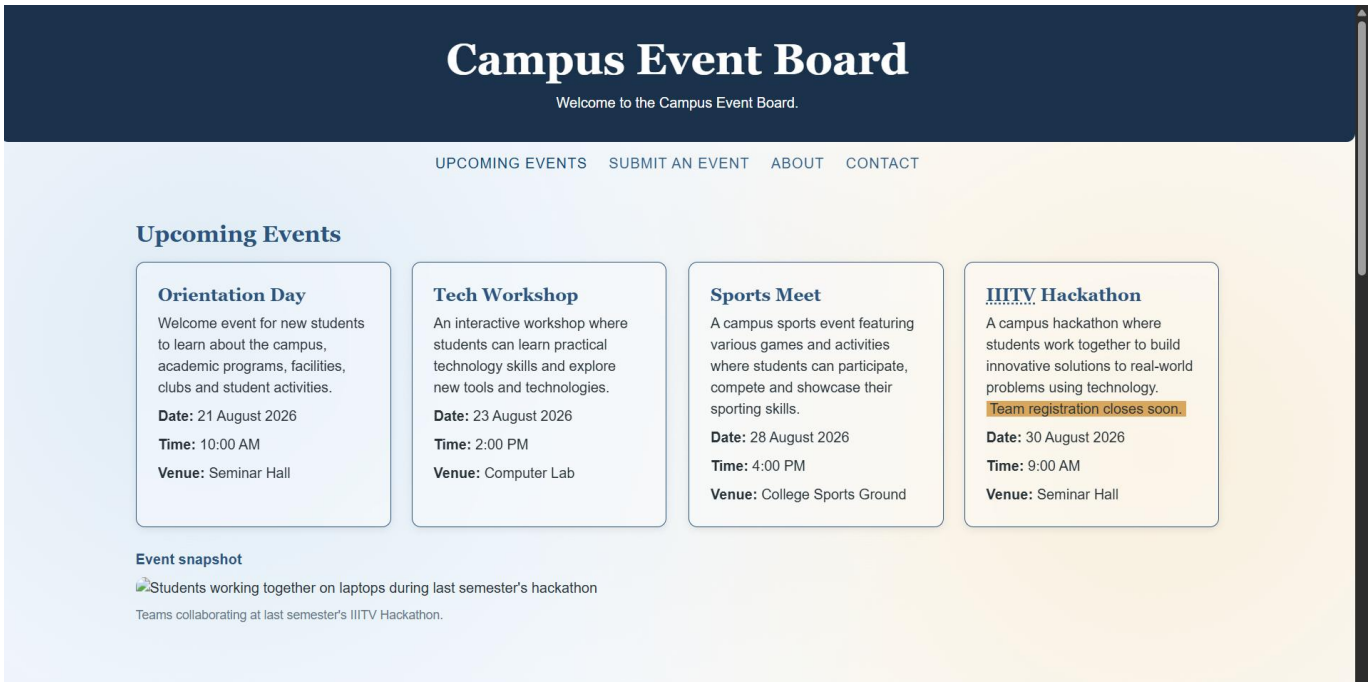
Type: clamp(), for example font-size: clamp(1.5rem, 4vw, 3rem); so the size responds to viewport width without a media query.

10. Prepare a GitHub repo for event-board and paste your final index.html and final style.css — the outcome of all the assignments. Give the repo link (and Pages link if live), and confirm it has a README and a readable commit history.

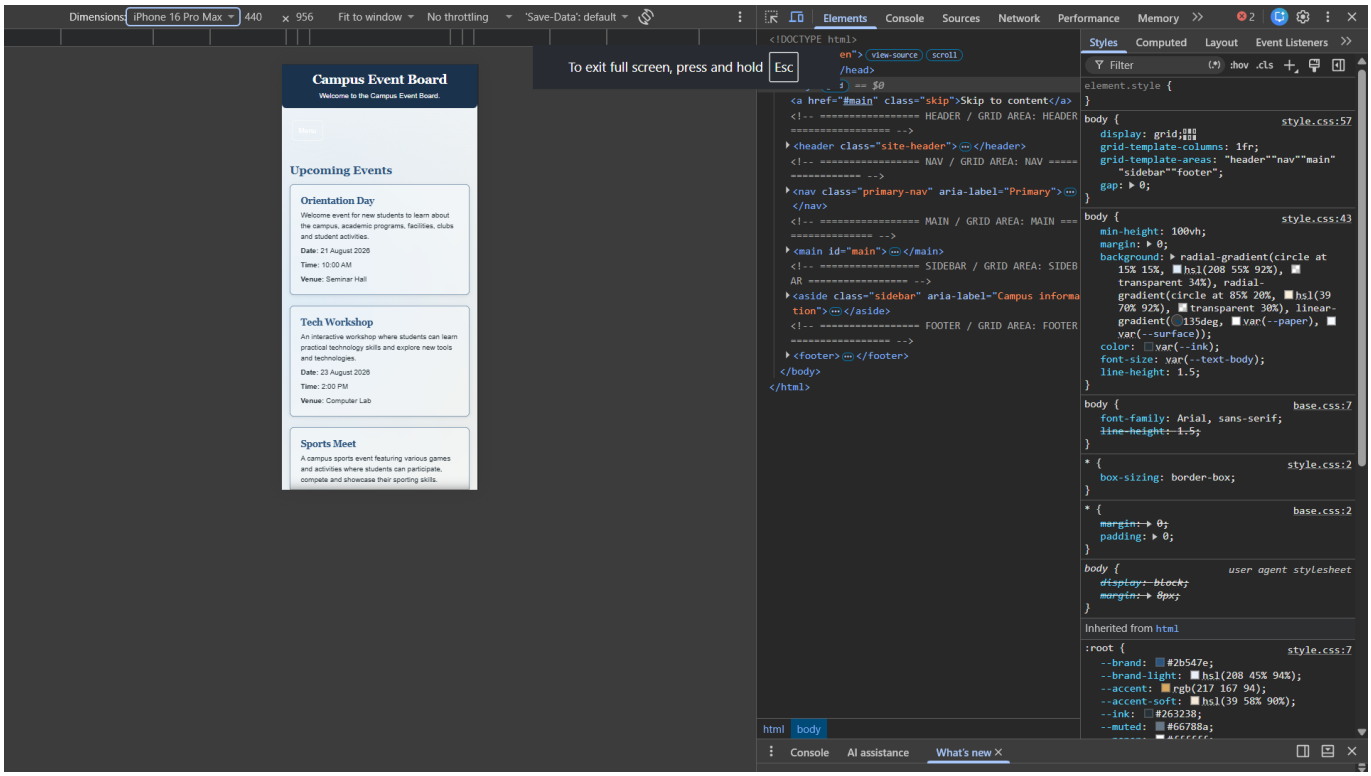
Repository name: event-board

GitHub repository link: [PASTE YOUR GITHUB REPOSITORY LINK HERE]
GitHub Pages link: [PASTE YOUR GITHUB PAGES LINK HERE, IF ENABLED]

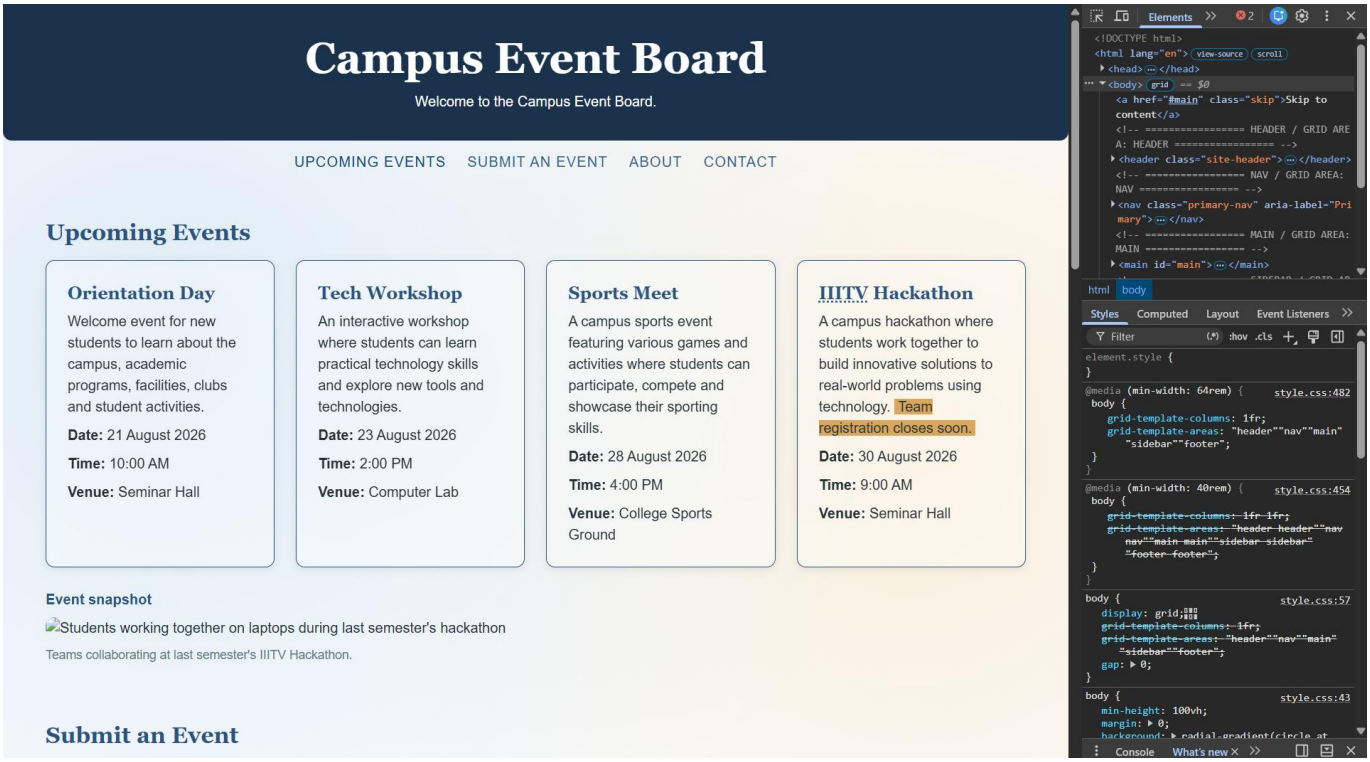
1. Styled Event Cards



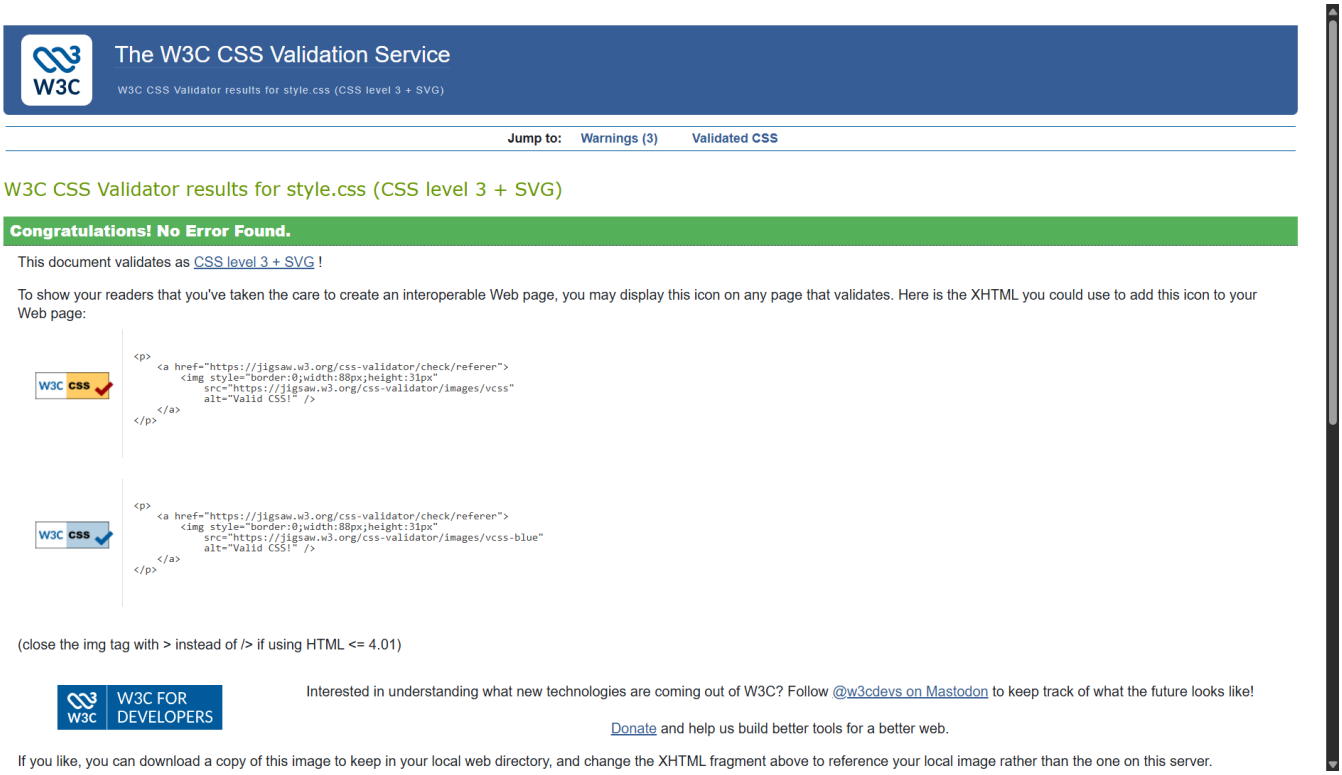
2. Narrow / Mobile Responsive Layout



3.Wide / Desktop Responsive Layout



4.W3C CSS Validator screenshot



Final Code Appendix

Final index.html

```
< <!DOCTYPE html>

<html lang="en">


<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1">

  <title>Campus Event Board - IIIT Vadodara</title>

  <meta name="description" content="Campus Event Board for IIIT Vadodara. Find upcoming events, submit your own event, and check event categories.">

  <link rel="icon" href="favicon.png">

  <meta property="og:title" content="Campus Event Board - IIIT Vadodara">

  <meta property="og:description" content="Find upcoming events, submit your own event, and stay updated with campus activities.">

  <meta property="og:image" content="og-image.png">


  <link rel="stylesheet" href="base.css">

  <link rel="stylesheet" href="style.css">

  <!-- Optional dark theme: load theme-dark.css after style.css when required. -->

</head>


<body>

  <a href="#main" class="skip">Skip to content</a>


  <!-- ================= HEADER / GRID AREA: HEADER ================= -->

  <header class="site-header">

    <h1>Campus Event Board</h1>
```

```
    <p>Welcome to the Campus Event Board.</p>
  </header>

  <!-- ================ NAV / GRID AREA: NAV ================ -->
  <nav class="primary-nav" aria-label="Primary">
    <!-- B6: hamburger is present for the future JavaScript interaction. -->
    <button class="menu-toggle" type="button" aria-expanded="false" aria-
controls="primary-links">Menu</button>
    <div class="nav-links" id="primary-links">
      <a href="#events" class="active" aria-current="page">Upcoming Events</a>
      <a href="#submit-event">Submit an Event</a>
      <a href="#about">About</a>
      <a href="#contact">Contact</a>
    </div>
  </nav>

  <!-- ================ MAIN / GRID AREA: MAIN ================ -->
  <main id="main">

    <!-- ================ UPCOMING EVENTS ================ -->
    <section id="events">
      <h2>Upcoming Events</h2>

      <!-- B3: cards reflow automatically with repeat(auto-fit, minmax()). -->
      <div class="event-grid">
        <article class="event">
          <h3>Orientation Day</h3>
          <p>Welcome event for new students to learn about the campus, academic
programs, facilities, clubs and student activities.</p>
```

```
          <p><strong>Date:</strong> <time datetime="2026-08-21">21 August
2026</time></p>

          <p><strong>Time:</strong> 10:00 AM</p>

          <p><strong>Venue:</strong> Seminar Hall</p>

        </article>



        <article class="event">

          <h3>Tech Workshop</h3>

          <p>An interactive workshop where students can learn practical technology
skills and explore new tools and technologies.</p>

          <p><strong>Date:</strong> <time datetime="2026-08-23">23 August
2026</time></p>

          <p><strong>Time:</strong> 2:00 PM</p>

          <p><strong>Venue:</strong> Computer Lab</p>

        </article>



        <article class="event">

          <h3>Sports Meet</h3>

          <p>A campus sports event featuring various games and activities where students
can participate, compete and showcase their sporting skills.</p>

          <p><strong>Date:</strong> <time datetime="2026-08-28">28 August
2026</time></p>

          <p><strong>Time:</strong> 4:00 PM</p>

          <p><strong>Venue:</strong> College Sports Ground</p>

        </article>



        <article class="event">

          <h3><abbr title="Indian Institute of Information Technology
Vadodara">IIITV</abbr> Hackathon</h3>

          <p>A campus hackathon where students work together to build innovative
solutions to real-world problems using technology. <mark>Team registration closes
soon.</mark></p>
```

```
          <p><strong>Date:</strong> <time datetime="2026-08-30">30 August
2026</time></p>

          <p><strong>Time:</strong> 9:00 AM</p>

          <p><strong>Venue:</strong> Seminar Hall</p>

        </article>

      </div>


      <figure>

        <img src="hackathon-photo.jpg" alt="Students working together on laptops during
last semester's hackathon">

        <figcaption>Teams collaborating at last semester's IIITV Hackathon.</figcaption>

      </figure>

    </section>


    <!-- ================= SUBMIT EVENT FORM ================= -->

    <section id="submit-event">

      <h2>Submit an Event</h2>

      <form action="#" method="post">

        <p>

          <label for="event-name">Event Name</label><br>

          <input type="text" id="event-name" name="event-name" required>

        </p>

        <p>

          <label for="event-date">Event Date</label><br>

          <input type="date" id="event-date" name="event-date" required>

        </p>

        <p>

          <label for="event-email">Organizer Email</label><br>

          <input type="email" id="event-email" name="event-email"
placeholder="name@iiitvadodara.ac.in" required>
```

```
        </p>
        <p>
          <label for="event-location">Event Location</label><br>
          <input type="text" id="event-location" name="event-location" required>
        </p>
        <p>
          <label for="event-description">Event Description</label><br>
          <textarea id="event-description" name="event-description" rows="5" cols="40"
placeholder="Briefly describe the event." required></textarea>
        </p>
        <button type="submit">Add Event</button>
      </form>
    </section>

    <!-- ================= ABOUT ================= -->
    <section id="about">
      <h2>About Campus Event Board</h2>
      <p><cite>Campus Event Board</cite> is a platform designed to help students stay
updated with events and activities happening at IIIT Vadodara.</p>
      <p>Students can discover coding competitions, hackathons, workshops, sports
events, cultural programs, seminars and other campus activities in one place.</p>
      <p>Visit the <a href="https://iiitvadodara.ac.in/">IIIT Vadodara website</a> for
institute updates.</p>
    </section>

    <!-- ================= EVENT CATEGORIES TABLE ================= -->
    <section id="event-info">
      <h2>Event Categories</h2>
      <table>
        <thead>
          <tr>
```

```
        <th>Event Category</th>
        <th>Typical Meeting Day</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>Coding Competitions</td>
        <td>Monday</td>
      </tr>
      <tr>
        <td>Workshops</td>
        <td>Wednesday</td>
      </tr>
      <tr>
        <td>Sports Events</td>
        <td>Friday</td>
      </tr>
      <tr>
        <td>Cultural Programs</td>
        <td>Saturday</td>
      </tr>
    </tbody>
  </table>
</section>

<!-- ================= CONTACT ================= -->
<section id="contact">
  <h2>Contact Us</h2>
  <p>Have an event that you want to add to the Campus Event Board?</p>
```

```
      <address>Email <a
href="mailto:events@iiitvadodara.ac.in">events@iiitvadodara.ac.in</a></address>

    </section>


  </main>


  <!-- ================= SIDEBAR / GRID AREA: SIDEBAR ================= -->

  <aside class="sidebar" aria-label="Campus information">

    <section id="announcements">

      <h2>Announcements</h2>

      <p>Check this page regularly for new campus events and announcements.</p>

      <p class="later-message">This message is hidden with display: none for a later
reveal.</p>

    </section>


    <section id="faq">

      <h2>Frequently Asked Questions</h2>

      <details>

        <summary>Who can submit an event?</summary>

        <p>Any registered student club or student can submit an event using the form
above.</p>

      </details>

      <details>

        <summary>How long does approval take?</summary>

        <p>Submitted events are usually reviewed and listed within two working days.</p>

      </details>

    </section>

  </aside>


  <!-- ================= FOOTER / GRID AREA: FOOTER ================= -->
```

```
  <footer>
    <nav aria-label="Footer">
      <a href="#events">Events</a>
      <a href="#about">About</a>
      <a href="#contact">Contact</a>
    </nav>
    <p>Campus Event Board &copy; 2026 | IIIT Vadodara</p>
  </footer>


</body>
</html>


>
```

Final style.css

```
/ /* Lab 5: the box-sizing reset is intentionally at the very top of style.css. */
* {
  box-sizing: border-box;
}


/* A2/A3: design-system palette. The palette includes hex, rgb(), hsl(), and a semi-transparent overlay. */
:root {
  --brand: #2b547e;
  --brand-light: hsl(208 45% 94%);
  --accent: rgb(217 167 94);
  --accent-soft: hsl(39 58% 90%);
  --ink: #263238;
  --muted: #66788a;
  --paper: #ffffff;
  --surface: #f7fbff;
```

```
  --border: #b7c8d6;
  --overlay: rgb(0 0 0 / 40%);
  --link: #164a78;
  --visited-link: #5c2d7d;
  --active-link: #9b2c2c;

  /* A5 type scale. */
  --text-sm: 0.875rem;
  --text-body: 1rem;
  --text-h3: 1.25rem;
  --text-h2: 1.6rem;
  --text-h1: clamp(1.75rem, 4vw, 3rem);

  --space-1: 0.25rem;
  --space-2: 0.5rem;
  --space-3: 0.75rem;
  --space-4: 1rem;
  --space-5: 1.5rem;
  --space-6: 2rem;
  --radius: 0.625rem;
}

/* A4: relative units, calc(), and a percentage/vw-based responsive layout. */
html {
  scroll-behavior: smooth;
}

body {
  min-height: 100vh;
```

```
  margin: 0;
  /* Glass backdrop: keeps the page visually rich without requiring an external image.
*/
  background:
    radial-gradient(circle at 15% 15%, hsl(208 55% 92%), transparent 34%),
    radial-gradient(circle at 85% 20%, hsl(39 70% 92%), transparent 30%),
    linear-gradient(135deg, var(--paper), var(--surface));
  color: var(--ink);
  font-size: var(--text-body);
  line-height: 1.5;
}

/* B2: mobile-first page grid. Every page region gets an explicit grid area. */
body {
  display: grid;
  grid-template-columns: 1fr;
  grid-template-areas:
    "header"
    "nav"
    "main"
    "sidebar"
    "footer";
  gap: 0;
}

h1,
h2,
h3 {
  color: var(--brand);
  font-family: Georgia, serif;
```

```
  line-height: 1.2;
}

h1 {
  margin: 0;
  font-size: var(--text-h1);
  text-align: center;
}

h2 {
  margin-bottom: var(--space-4);
  font-size: var(--text-h2);
}

h3 {
  margin-bottom: var(--space-2);
  font-size: var(--text-h3);
}

p {
  max-width: 65ch;
}

small,
figcaption {
  font-size: var(--text-sm);
}

/* B2: header occupies the header grid region. */
```

```
.site-header {
  grid-area: header;
  position: relative;
  overflow: hidden;
  padding: clamp(1rem, 3vw, 2rem) 5%;
  background-color: var(--brand);
  border-radius: 0 0 var(--radius) var(--radius);
  color: var(--paper);
}

/* A3: semi-transparent overlay, useful for a banner-style header. */
.site-header::before {
  content: "";
  position: absolute;
  inset: 0;
  background-color: var(--overlay);
  pointer-events: none;
}

.site-header > * {
  position: relative;
  z-index: 1;
}

.site-header h1,
.site-header p {
  color: var(--paper);
  margin-inline: auto;
}
```

```
.site-header p {
  margin-top: var(--space-2);
  text-align: center;
}

/* B6: hamburger is visible on phones; JavaScript can control it later. */
.menu-toggle {
  display: inline-block;
  margin-top: var(--space-4);
  padding: 0.6rem 0.9rem;
  border: 1px solid var(--paper);
  border-radius: 0.4rem;
  background: transparent;
  color: var(--paper);
  cursor: pointer;
}

/* B6: full navigation is deliberately hidden on phones. */
.nav-links {
  display: none;
  flex-wrap: wrap;
  justify-content: center;
  gap: var(--space-3) var(--space-5);
}

/* B2/B6: primary navigation is its own grid region. */
.primary-nav {
  grid-area: nav;
```

```
  width: min(90%, 60rem);
  margin: 0 auto;
  padding: var(--space-3) 0;
}

.nav-links a {
  text-decoration: none;
  text-transform: uppercase;
  letter-spacing: 0.06em;
}

/* :not() keeps the active page link visually distinct. */
.nav-links a:not(.active) {
  opacity: 0.88;
}

/* A4: width uses calc() while the container stays capped for readable lines. */
.page-container {
  width: min(90%, 60rem);
  margin: 0 auto;
}

/* B2: main is a named grid region. */
main {
  grid-area: main;
  width: min(92%, 76rem);
  margin: 0 auto;
  padding: var(--space-5) 0;
}
```

```
main > section {
  margin: 0 0 var(--space-6);
  padding: var(--space-4) 0;
}

/* B3: responsive card grid; auto-fit + minmax reflows without a media query. */
.event-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(15rem, 1fr));
  gap: var(--space-5);
}

/* A8 + visual enhancement: each event keeps the required 1px border, radius, shadow,
padding and margin while adding a frosted-glass surface. */
.event {
  margin: 0;
  padding: var(--space-5);
  border: 1px solid var(--brand);
  border-radius: var(--radius);
  background: rgb(255 255 255 / 35%);
  -webkit-backdrop-filter: blur(12px);
  backdrop-filter: blur(12px);
  box-shadow: 0 0.125rem 0.5rem rgb(0 0 0 / 8%);
}

.event p {
  margin-top: var(--space-2);
}
```

```
.event mark {
  background-color: var(--accent);
  color: var(--ink);
  padding-inline: var(--space-1);
}

/* A9: responsive images plus object-fit for a consistent thumbnail/banner shape. */
figure {
  margin: var(--space-5) 0 0;
}

figure img {
  display: block;
  width: 100%;
  max-width: 100%;
  height: auto;
  max-height: 22rem;
  object-fit: cover;
  border-radius: var(--radius);
}

figure::before {
  content: "Event snapshot";
  display: block;
  margin-bottom: var(--space-2);
  color: var(--brand);
  font-weight: bold;
}
```

```
figcaption {
  margin-top: var(--space-2);
  color: var(--muted);
}

/* A4: 100%-based width plus calc() for the form controls. */
input,
textarea {
  width: min(100%, calc(100% - 1rem));
  max-width: 30rem;
  padding: var(--space-3);
  border: 1px solid var(--border);
  border-radius: 0.4rem;
  background-color: var(--surface);
  color: var(--ink);
  font: inherit;
  line-height: inherit;
}

[type="email"],
[type="date"] {
  border-color: var(--brand);
}

input::placeholder,
textarea::placeholder {
  color: var(--muted);
  font-style: italic;
}
```

```
input:focus,
textarea:focus {
  outline: 0.1875rem solid var(--accent);
  outline-offset: 0.125rem;
}

button {
  font: inherit;
}

button[type="submit"] {
  padding: var(--space-3) var(--space-5);
  border: 0;
  border-radius: 0.4rem;
  background-color: var(--brand);
  color: var(--paper);
  cursor: pointer;
}

/* B4: table uses full available width and striped rows. */
table {
  width: 100%;
  border-collapse: collapse;
}

th,
td {
  padding: var(--space-3);
```

```
  border: 1px solid var(--border);
  text-align: left;
}

thead {
  background-color: var(--brand);
  color: var(--paper);
}

tbody tr:nth-child(even) {
  background: rgb(255 255 255 / 35%);
  -webkit-backdrop-filter: blur(12px);
  backdrop-filter: blur(12px);
}

/* A3: accent hover state also demonstrates rgb()/opacity-style transparency. */
a:hover {
  color: var(--ink);
  text-decoration: underline;
}

/* LVHA link states. */
a:link {
  color: var(--link);
}

a:visited {
  color: var(--visited-link);
}
```

```
a:active {
  color: var(--active-link);
}

a:focus {
  outline: 0.1875rem solid var(--accent);
  outline-offset: 0.1875rem;
}

a[href^="https"]::after {
  content: " ↗";
  font-size: 0.85em;
}

/* A9: deliberate display use; this utility is hidden until needed. */
.later-message {
  display: none;
}

/* A6/A7: the skip link demonstrates positioned box-model spacing. */
.skip {
  position: absolute;
  left: -999rem;
  top: 0;
  padding: var(--space-3);
  background-color: var(--ink);
  color: var(--paper);
}
```

```
.skip:focus {
  left: var(--space-4);
  top: var(--space-4);
}

/* B5: sidebar is a separate region; it moves beside main at large widths. */
.sidebar {
  grid-area: sidebar;
  width: min(92%, 76rem);
  margin: 0 auto;
  padding: var(--space-5) 0;
}

.sidebar section {
  margin-bottom: var(--space-5);
  padding: var(--space-4);
  border: 1px solid var(--border);
  border-radius: var(--radius);
  background: rgb(255 255 255 / 35%);
  -webkit-backdrop-filter: blur(12px);
  backdrop-filter: blur(12px);
  box-shadow: 0 0.125rem 0.5rem rgb(0 0 0 / 8%);
}

/* Keep the announcement and FAQ panels aligned and prevent grid overflow. */
.sidebar {
  align-self: start;
  min-width: 0;
```

```
}

.sidebar section h2 {
  margin-top: 0;
  margin-bottom: var(--space-3);
}

.sidebar section p {
  margin-top: var(--space-2);
}

#faq summary {
  padding: var(--space-2) 0;
  cursor: pointer;
  font-weight: 600;
}

#faq details p {
  margin: var(--space-2) 0 0;
}

details {
  padding: var(--space-3) 0;
  border-bottom: 1px solid var(--border);
}

details:last-child {
  border-bottom: 0;
}
```

```
address {
  font-style: initial;
}

/* B2: footer occupies the footer grid region. */
footer {
  grid-area: footer;
  margin-top: var(--space-6);
  padding: var(--space-5) 5%;
  background-color: var(--brand-light);
  text-align: center;
}

footer nav {
  display: flex;
  flex-wrap: wrap;
  justify-content: center;
  gap: var(--space-4);
}

footer p
{
  width: 100%;
  max-width: none;
  text-align: center;
  margin: 0.5rem 0 0;
}
```

```
/* B5: first breakpoint adds a second page-grid track. */
@media (min-width: 40rem) {
  body {
    grid-template-columns: 1fr 1fr;
    grid-template-areas:
      "header header"
      "nav nav"
      "main main"
      "sidebar sidebar"
      "footer footer";
  }

  main {
    padding-inline: var(--space-4);
  }

  .sidebar {
    width: min(92%, 76rem);
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: var(--space-5);
  }

  .sidebar section {
    margin: 0;
  }
}

/* B5/B6: desktop keeps main and sidebar in a clean vertical composition, while the nav
becomes a full link row. */
```

```
@media (min-width: 64rem) {
  body {
    grid-template-columns: 1fr;
    grid-template-areas:
      "header"
      "nav"
      "main"
      "sidebar"
      "footer";
  }

  .menu-toggle {
    display: none;
  }

  .nav-links {
    display: flex;
  }

  main {
    width: min(92%, 76rem);
    padding-inline: 0;
  }

  .sidebar {
    width: min(92%, 76rem);
    margin: 0 auto;
    padding: 0 0 var(--space-5);
    display: grid;
```

```
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: var(--space-5);
  }

  .sidebar section {
    margin-bottom: 0;
  }
}

/
```